Southern New Hampshire University

DriverPass System Design Document
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CS 255
Systems Analysis and Design

UML Diagrams

UML Use Case Diagram

This diagram shows who uses the DriverPass system and what each person can do in it. I grouped it around what Liam and Ian asked for in the interview.

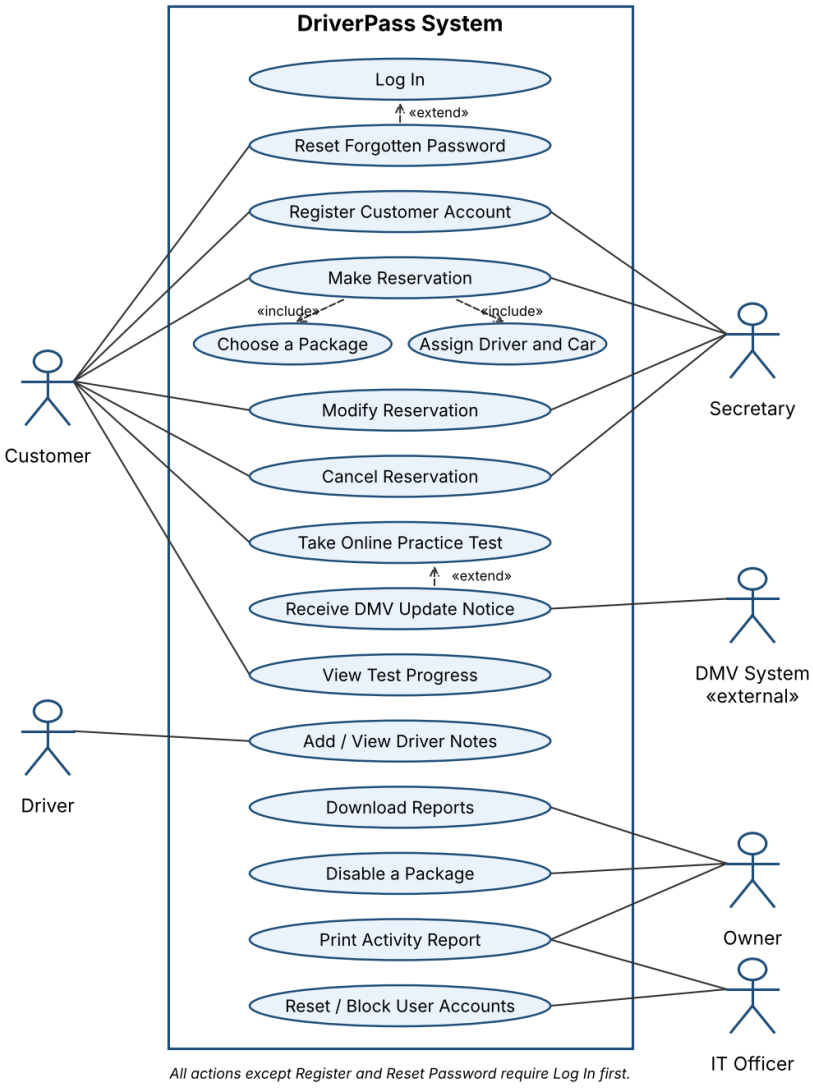


Figure 1. Use case diagram for the DriverPass system

- Customers can reset a forgotten password on their own, book, change, or cancel a 2-hour lesson online, take practice tests, and see their test progress.
- Secretary can register customers and book, change, or cancel lessons for people who call or walk in.
- Drivers add comments about each lesson and can look at their notes.
- Owner can disable a package, download reports to use in Excel, and print an activity report.
- IT officer can reset or block any user account, and can also print the activity report.
- DMV system is outside the company. It sends an update when rules or sample questions change, and that update feeds into the practice tests.
- Making a reservation includes choosing a package and assigning a driver and car, so those two always happen as part of booking. Every action (except registering and resetting a password) needs the person to log in first, which is how each role only sees what it should.

UML Activity Diagrams

I picked two use cases: Make Reservation and Reset Forgotten Password. Both came straight from the interview.

Activity Diagram 1: Make Reservation

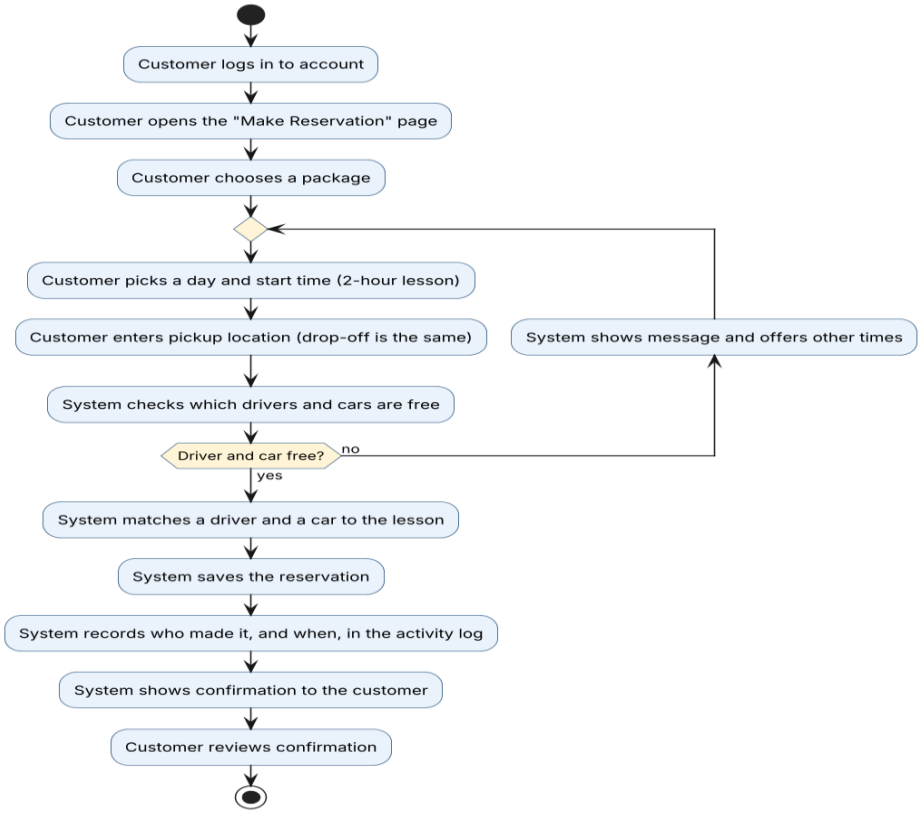


Figure 2. Activity diagram for Make Reservation

- The customer logs in, picks a package, then picks a day and start time and enters a pickup spot (drop-off is the same). The system checks for a free driver and car. If none are free, it offers other times and the customer tries again. Once there's an opening, the system matches a driver and car, saves the reservation, records who made it in the activity log, and shows a confirmation. This covers Liam's need to know which student has which driver, time, and car, and who made the booking.

Activity Diagram 2: Reset Forgotten Password

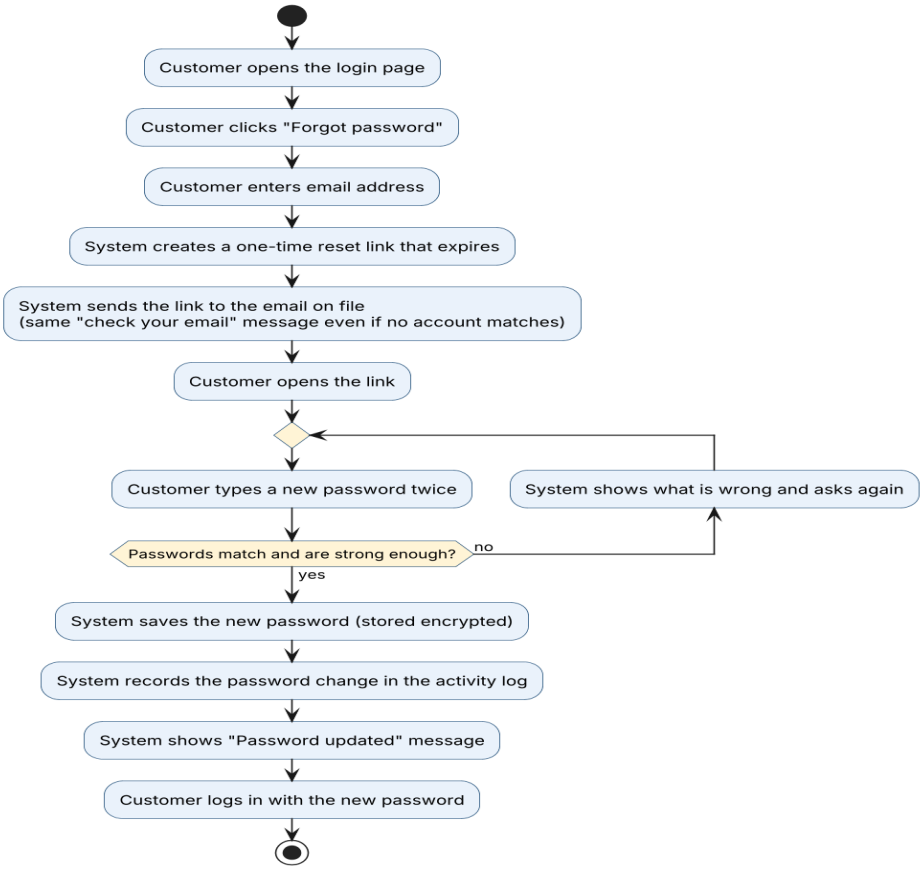


Figure 3. Activity diagram for Reset Forgotten Password

- The customer clicks "Forgot password" and enters their email. The system emails a one-time link that expires. The customer opens it and types a new password twice. If the two don't match or it's too weak, the system asks again. If it's good, the new password is saved (encrypted) and the change goes in the activity log. This is what Ian asked for: customers can reset their own password without calling him. The system shows the same message whether or not the email has an account, so nobody can guess who's a customer.

UML Sequence Diagram

This sequence diagram follows the Make Reservation use case from the customer's click to the confirmation.

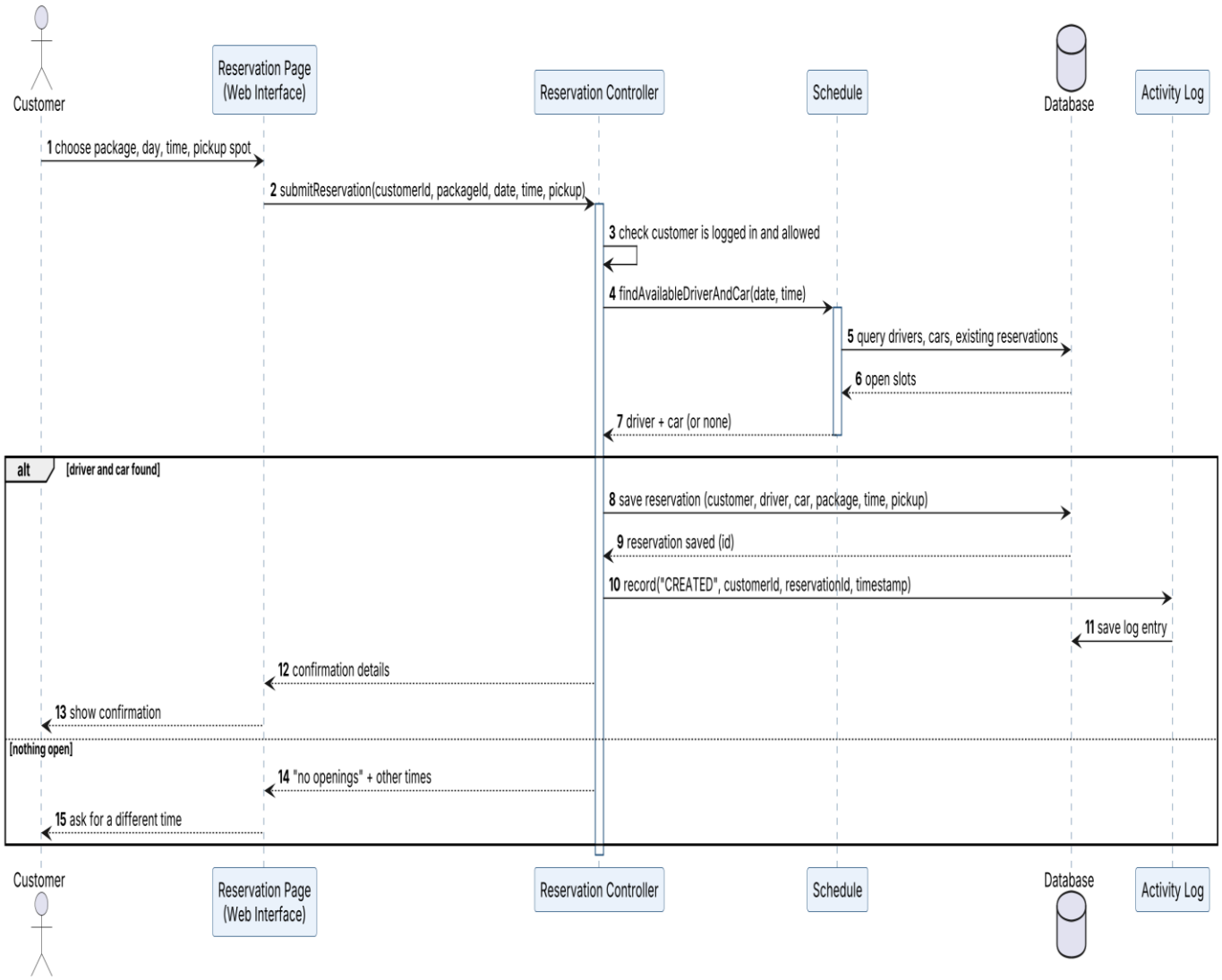


Figure 4. Sequence diagram for Make Reservation

- The customer fills in the reservation page and submits it. The reservation controller checks the customer is logged in and allowed, then asks the schedule to find an open driver and car. The schedule looks in the database and answers. If there's an opening, the controller saves the reservation, writes an entry to the activity log (who, what, when), and sends the confirmation back to the page. If not, the customer gets other times to pick from. Keeping the log step in the flow is how the system tracks who made, changed, or canceled each reservation.

UML Class Diagram

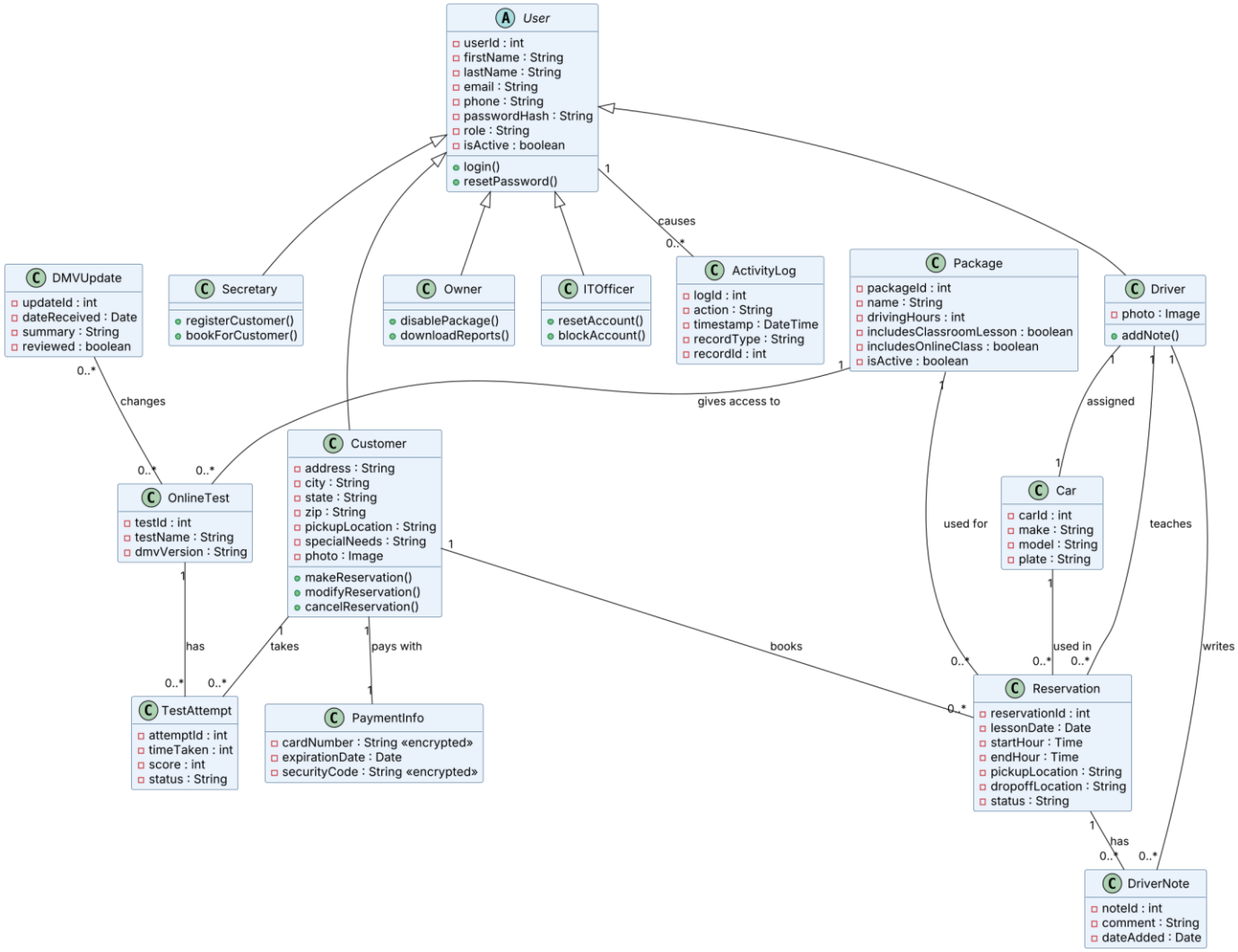


Figure 5. Class diagram for the DriverPass system

- This class diagram shows the main pieces of data in the system and how they connect. Methods are included for the user roles only. Everything else just shows attributes.
- User is the parent class (name, email, password, role, active or blocked). Customer, Secretary, Driver, Owner, and ITOfficer inherit from it, so they all log in the same way but have different rights.
- Reservation links one customer, one driver, one car, and one package to a 2-hour lesson with a pickup and drop-off location. A Driver has one Car and many Reservations.
- Package has an isActive flag so Liam can disable a package without deleting it. It also records the hours and whether it includes the classroom lesson and online class.
- OnlineTest and TestAttempt store the test name, time taken, score, and status (not taken, in progress, failed, passed) for the progress screen.
- DriverNote holds a driver's comments for a lesson, and shows up in the driver notes table.
- DMVUpdate records each notice from the DMV so tests can be kept current.
- ActivityLog records every action, who did it, and when. The printed activity report comes from this.
- PaymentInfo keeps the card number, expiration date, and security code, marked as encrypted.

Technical Requirements

These requirements come from what Liam and Ian said in the interview, the use cases above, and the functional and nonfunctional requirements from the business requirements document.

Hardware

- No servers on site. Ian said they don't want to deal with backups and security, so the system runs on cloud servers (a web/application server and a database server) that the hosting company manages.
- Staff and customers use what they already have: a desktop, laptop, tablet, or phone with an up-to-date web browser and an internet connection.
- A printer at the office for the activity report.

Software

- A web application with a responsive layout, so it works on computers and mobile devices (HTML, CSS, JavaScript on the front end).
- A server-side application that handles logins, roles, reservations, scheduling, and the activity log. The developers can pick the language and framework (for example Java or Python).
- A relational database (like PostgreSQL or MySQL) for users, packages, reservations, cars, tests, notes, and logs.
- Export to CSV or Excel files so Liam can download reports and work on them at home.
- An email service to send password reset links and booking confirmations.
- A connection to the DMV (an API or data feed) that sends a notification when rules, policies, or sample questions change.
- A payment processor so card details are handled safely instead of being stored in plain text.

Tools

- Lucidchart for the UML diagrams, plus a version control system (Git) for the code.
- A testing tool or framework for the Test System step in the project schedule, and a bug or task tracker for the team.

Infrastructure

- Cloud hosting with automatic daily backups and a way to restore them.
- HTTPS (encrypted connections) for every page.
- Role-based access: Owner, IT officer, Secretary, Driver, and Customer each see only what their role allows. The IT officer can reset or block any account.
- Online and offline use: data can be changed only while online, so there aren't duplicate copies on different servers. Offline, users can open reports and downloaded files, but not edit records.
- The system has to handle 10 cars and drivers and the customer load for one company without slowing down. It should be up during business hours and most evenings, since customers book and study at night.
- Passwords stored encrypted (hashed), and card number and security code encrypted. Every create, change, and cancel is logged with the user and time.